Entanglement Swapping Across Architectures

Minimal Cross-Stack Quantum Benchmark

Scope: Pure Python · QuTiP · Qiskit · IonQ-native · Quantinuum-native · Strawberry Fields CV.

Purpose: Show how the same entanglement-swapping primitive can be expressed across abstraction layers and paradigms using identical DV metrics and CV-native generalized metrics for transparent cross-architecture comparison.

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Why this matters

Entanglement swapping is a foundational primitive in quantum technology. It enables two remote systems to become entangled without direct interaction through intermediate measurement and conditioning, making it central to quantum communication, repeater concepts, modular quantum computing, and distributed sensing. Because it combines state preparation, measurement, and coordination in a compact protocol, it is also a useful benchmark for comparing how different quantum software ecosystems express the same operational task.

What was built

This project implements a minimal local-simulation benchmark for entanglement swapping across six software stacks:

- pure Python
- QuTiP
- Qiskit
- IonQ-native syntax
- Quantinuum-native compilation flow
- Strawberry Fields (CV)

The design goal is simplicity. The physics is kept intentionally compact so the emphasis stays on syntax, abstraction, and cross-architecture translation rather than on algorithmic complexity.

Benchmark design

For the **discrete-variable (DV)** implementations, the protocol is identical:

- prepare two Bell pairs
- perform Bell-state projection on the middle qubits
- evaluate the swapped state on the end qubits

All DV stacks report the same exact metrics:

- **success probability**
- **Bell-state fidelity**
- **entanglement yield = success probability × Bell fidelity**

Because the DV implementations are ideal and local, the results should match exactly up to numerical precision.

For the **continuous-variable (CV)** implementation, the protocol is preserved in the same operational spirit, but the metric is not artificially forced to match the DV case. Instead, the Strawberry Fields version uses entangled Gaussian resources, intermediate interference, and homodyne measurement, and reports:

- **success probability**
- **native entanglement metric**
- **generalized entanglement yield**

$\boxed{\textbf{DV metrics are identical; CV metrics are generalized but operationally aligned.}}$

Result

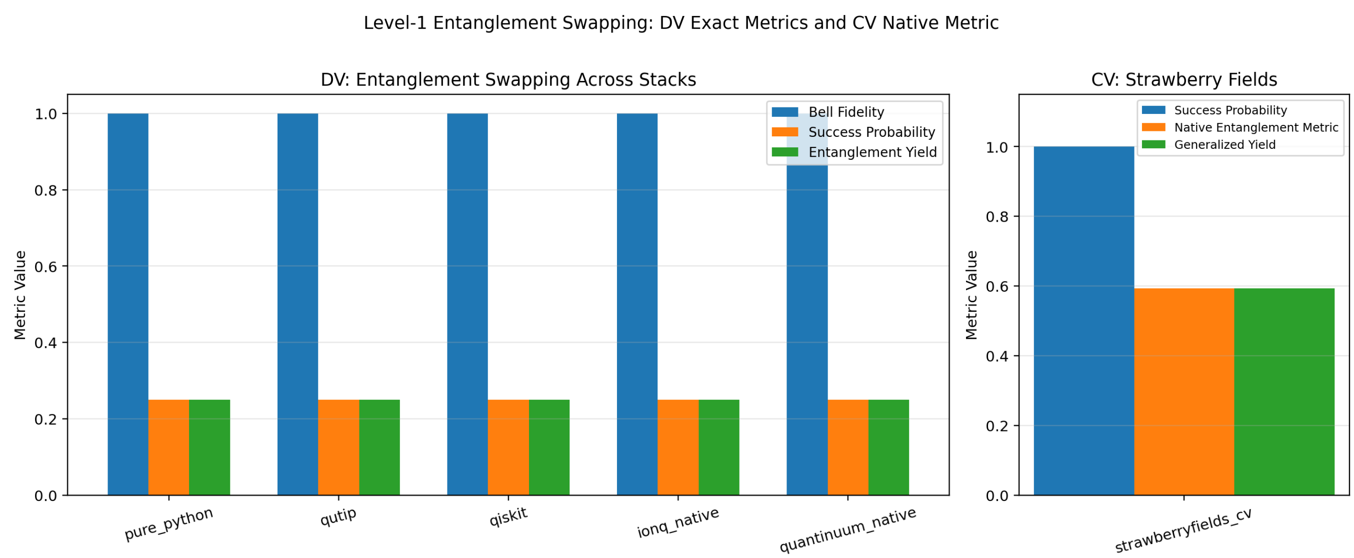

Figure 1: **Two-panel comparison for minimal entanglement swapping across architectures.** Left: exact-metric DV comparison across pure Python, QuTiP, Qiskit, IonQ-native, and Quantinuum-native flows. Right: Strawberry Fields CV implementation shown separately using a native entanglement metric and generalized yield.

Why this artifact is useful

This benchmark is best understood as a **cross-architecture translation artifact**. It shows how one compact quantum-networking primitive can be expressed and evaluated consistently across:

- explicit linear algebra
- low-level physics simulation
- circuit-based SDKs
- native vendor-oriented syntax
- continuous-variable photonic modeling

The result is a small but rigorous example of architecture-aware benchmarking, local reproducibility, and transparent quantum-technology communication.

Key takeaway

This project holds the protocol fixed while translating it across abstraction layers and paradigms. The DV side demonstrates exact metric consistency across multiple stacks, while the CV side extends the same operational idea using architecture-native measurement and entanglement metrics. The result is both a technical benchmark and a communication-ready artifact for cross-platform quantum systems work.

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